



Trophoblast differentiation and embryonic development are delayed in lactating cows compared to nulliparous heifers

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ABSTRACT

The objective of this study was to evaluate the effects of parity on early embryonic development and placentation in Holstein cattle. Pregnant cows were classified as nulliparous ($n = 16$), primiparous ($n = 7$), or multiparous ($n = 8$) after estrous synchronization for timed artificial insemination (d 0 = day of timed AI). Pregnancy was confirmed on d 28 after AI via transrectal ultrasonography. Cervical epithelial cells were collected daily from d 14 to 20 to assess ISG15 and MX2 mRNA abundance. Blood samples were collected on d 0, 4.5, 7, 10, and 12 and daily from d 14 to 30; additional samples were collected every 12 h from d 18 to d 24 to assess circulating pregnancy-associated glycoproteins (PAG) concentration. Data were analyzed using mixed-effects models for repeated measures, one-way ANOVA, and Pearson correlation, with Tukey-adjusted comparisons and significance set at $P < 0.05$. There was only an effect of time for MX2 abundance peaking on d17 and decreasing thereafter. For ISG15, mRNA abundance was greater in primiparous and multiparous compared with nulliparous on d 17, and greater in primiparous than nulliparous on d 18. Concentrations of PAG started to increase earlier in nulliparous heifers ($d\ 18.9 \pm 0.1$; $\pm$ SEM) compared with primiparous ($d\ 20.0 \pm 0.3$) and multiparous cows ($d\ 20.6 \pm 0.3$). Overall, PAG was greater in nulliparous than primiparous from d 20 to 21.5 and in nulliparous than multiparous from d 19.5 to 22. Concentrations of progesterone (P4) were greater in nulliparous heifers from d 4.5 to 12, and greater P4 concentration was associated with earlier increase in PAG. In conclusion, this study demonstrates that parity influences early embryonic development, with earlier trophoblast differentiation and acqui-

sition of PAG secretory capacity in nulliparous compared with primiparous and multiparous lactating cows.

Key words: embryonic development, pregnancy-associated glycoproteins, interferon-stimulated genes, progesterone

INTRODUCTION

Reproductive efficiency is a critical factor influencing both the productivity and profitability of dairy operations. Among the primary causes of reduced efficiency, pregnancy loss in lactating dairy cows represents a major economic burden due to increased rebreeding, extended calving intervals, lower milk yield, and higher culling rates (De Vries, 2006; Speckhart et al., 2023). Notably, pregnancy loss is positively associated with parity, with lactating cows—particularly multiparous cows—having substantially greater losses than nulliparous heifers (Fernandez-Novo et al., 2020). Most of these losses occur in the first month of gestation when important biological processes occur, such as embryonic elongation, maternal recognition of pregnancy, and implantation (Diskin and Morris, 2008; Wiltbank et al., 2016). These events involve dynamic interactions between the conceptus and the maternal system, and any disruptions during this critical period may lead to pregnancy failure (Spencer et al., 2004a).

Early stages of gestation in cattle is characterized by rapid morphological and functional changes in the developing embryo and by dynamic adaptations of the maternal reproductive tract required to support conceptus growth (Sánchez et al., 2019). Following fertilization, the embryo progresses to the blastocyst stage by 7 d after fertilization (Rizos et al., 2002; Prados et al., 2012). After hatching from the zona pellucida, the embryo differentiates into an ovoid and then filamentous structure, entering a phase of rapid elongation from d 16 to 20 after insemination (Blomberg et al., 2008). During this period, the developing conceptus becomes increasingly depen-

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dent on secretions from the maternal endometrium to sustain its growth and differentiation (Gray et al., 2001). Uterine endometrial glands secrete histotroph composed of nutrients, growth factors, cytokines, and signaling molecules that supports conceptus elongation, trophoblast differentiation, implantation, and early placental development (Gray et al., 2001; Bazer et al., 2011). The composition and secretion of histotroph are regulated specially by progesterone (P4), which prepares the endometrium to sustain conceptus elongation (Spencer et al., 2016). Therefore, alterations in the maternal P4 dynamics during early pregnancy can modify uterine function and influence embryonic development.

During elongation, the trophoblast produces interferon-tau (IFNT) that acts locally on the endometrium to suppress PGF_{2α} secretion, preventing luteolysis and supporting continued P4 production (Bazer et al., 1997; Spencer et al., 2004b; Hansen et al., 2017). The secretion of IFNT peaks around d 17 and stimulates the expression of interferon-stimulated genes (ISG) in maternal tissues, including the peripheral blood leukocytes, endometrium, cervix, corpus luteum, liver, and hypothalamus (Oliveira et al., 2008; Hansen et al., 2017; Domingues et al., 2024). As the conceptus continues to develop, mononuclear trophoblast cells differentiate and fuse with uterine epithelial cells around d 20, forming bi- and multinucleated trophoblast giant cells that initiate placentation and secrete pregnancy-associated glycoproteins (PAG; Wooding, 1992, 2022; Wooding et al., 1996; Seo et al., 2023). As the embryonic trophoblast fuses with uterine epithelial cells, IFNT secretion declines (Guillomot et al., 1990; Sakurai et al., 2012). Therefore, the transition from IFNT-driven signaling to PAG secretion is associated with the initial structural and functional establishment of the synepitheliochorial placenta that is essential for sustaining pregnancy in ruminants.

Although it is well established that parity affects reproductive performance—heifers consistently have higher fertilization success, more viable embryos, and lower pregnancy loss than lactating cows (Pursley et al., 1997; Sartori et al., 2002; Fernandez-Novoa et al., 2020)—emerging evidence suggests that these differences extend to conceptus–maternal interactions during early pregnancy. Middleton and Pursley (2019) reported an earlier increase in circulating concentrations of PAG compared with lactating cows, suggesting an earlier initiation of conceptus attachment. Similarly, Santos et al. (2023) demonstrated that multiparous cows experience a delayed increase in circulating PAG concentrations compared with primiparous cows, which was associated with a greater risk of pregnancy loss. These findings highlight that parity may influence not only fertilization and embryo viability but also the timing and progression of peri-implantation events.

Despite these advancements, key gaps remain. It is unknown whether parity affects (1) the magnitude or temporal dynamics of conceptus-derived IFNT signaling during early pregnancy, as reflected by maternal ISG expression, and (2) whether the delayed increase in circulating PAG concentrations observed in multiparous cows also reflects a slower rate of embryonic development after attachment begins. Moreover, whether differences in maternal P4 profiles contribute to parity-associated variation in conceptus–maternal signaling has not been clearly defined. Addressing these questions is essential to understand why multiparous cows exhibit higher early pregnancy loss and to identify potential physiological targets for improving fertility in high-producing herds.

Therefore, the objective of this study was to determine the effect of parity on peri-implantation embryonic development in dairy cattle by characterizing (1) IFNT signaling through the expression of ISG in cervical epithelial cells, (2) early placental development through circulating PAG concentrations, and (3) luteal support through circulating P4 concentrations. We hypothesized that nulliparous heifers have (1) earlier and greater IFNT signaling as indicated by the expression of ISG, and (2) greater and faster increase in circulating PAG concentrations compared with primiparous and multiparous cows during early pregnancy.

MATERIALS AND METHODS

Ethics Statement

All animal procedures were performed in accordance with the *Guide for the Care and Use of Agricultural Animals in Research and Teaching* (ADSA et al., 2020) and were approved by the Institutional Animal Care and Use Committee of the University of Wisconsin–Madison (Protocol ID: A006603-R01).

Experimental Design and Animal Management

The study was conducted at the US Dairy Forage Research Center (Prairie du Sac, WI) from September 2022 to May 2023. A total of 68 Holstein cows were enrolled, including 34 heifers (average age = 15.4 ± 3 mo, $\pm$ SEM), 11 primiparous cows, and 22 multiparous cows. The average DIM at enrollment for lactating cows was 59 ± 4.2 d, and the average milk yield was 46.7 ± 3.3 kg/d. Only heifers bred for the first time and cows bred for the first service were included in the study. Frozen conventional semen from 4 bulls purchased by the farm were used for insemination.

All animals were subjected to the same synchronization of ovulation protocol before timed artificial insemination (TAI). The day of TAI was considered d 0.

The protocol was a modification of the G6G/Ovsynch protocol (Bello et al., 2006). As part of the presynchronization, animals received PGF_{2α} (0.5 mg of cloprostenol; Estroplan, Parnell) on d -17 and d -16 and GnRH (0.1 mg; Ganabreed; Parnell) on d -15. The modified breeding Ovsynch started on d -9 with another GnRH and the insertion of a controlled internal drug release (CIDR; Zoetis) that was removed 6 d later (d -3). The protocol was finalized with PGF_{2α} on d -3 and d -2 and the final GnRH 32 h after the last PGF_{2α} (d -1). Transrectal ultrasonography was performed every 12 h for 48 h following the final GnRH to verify ovulation which was confirmed by the disappearance of the dominant follicle. Only cows that ovulated from TAI to 24 h later were included in the study (n = 67).

Pregnancy status was assessed on d 28 post-TAI using transrectal ultrasonography to detect embryonic heart-beat. Only cows that maintained pregnancy through d 56 were included in the final analysis. Pregnant cows were classified according to parity for further analysis: nulliparous (parity = 0; n = 16), primiparous (parity = 1; n = 7), and multiparous (parity ≥ 2; n = 8).

Blood Collection for P4 and PAG Concentration in the Maternal Circulation

Blood collection was performed on d 0, 4.5, and 7, and daily from d 14 to 30 after artificial insemination (AI). To more precisely determine the timing of circulating PAG increase, blood samples were collected every 12 h from d 18 to 24. Blood samples were collected from the coccygeal vein using EDTA-coated tubes, immediately placed on ice, and centrifuged at $1,800 \times g$ for 10 min at 4°C to isolate plasma. The plasma was then stored at -20°C until analysis of P4 and PAG. Progesterone concentrations were measured using a solid-phase radioimmunoassay kit (ImmuChem Coated Tube P4 ¹²⁵I RIA Kit; MP Biomedicals, Solon, OH) with antibody-coated tubes and ¹²⁵I-labeled P4. The assay sensitivity was 0.053 ng/mL with intra-assay CV of 12%. For measurement concentration of PAG, samples were sent to BioTracking (BioPRYN Flex ELISA assay, Moscow, ID). The day of PAG increase for each cow was defined as the first day with a ≥10% increase above the baseline (calculated as the average of d 14, 15, 16, and 17) followed by further increase for at least 2 consecutive days. The day that cows reached 8 ng/mL (detection limit for the assay) of PAG was also defined for each cow.

Collection of Cervical Epithelial Cells and Gene Expression Analysis

To assess the expression of ISG15 and MX2 mRNA, cervical epithelial samples were obtained daily from d

14 to 20 after TAI following the methodology described by Domingues et al. (2024). Cervical sampling was performed using a cytology brush mounted to a bovine AI gun covered with a disposable sheath to prevent contamination with other cell types. The AI gun was guided to the cervical os, and the brush was extended approximately 1 cm into the cervical canal. The brush was exposed, and the gun was gently rotated to collect cells. After sample collection, the brush was immediately placed on dry ice and subsequently stored at -80°C until RNA extraction. To extract RNA from cervical samples, TRIzol reagent (Invitrogen, Pleasanton, CA) was used according to the protocol provided by the manufacture. For cDNA synthesis, 0.5 µg of RNA was reverse-transcribed using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Thermo Fisher Scientific, Waltham, MA). The synthesized cDNA was diluted 1:2 and used in quantitative real-time PCR (qRT-PCR) to quantify ISG15 and MX2 mRNA abundance. Primers were designed for bovine ACTB (forward: 5'-acttgccgacagaaacagat-3'; reverse: 5'-caccttcaccgttcagttt-3'), ISG15 (forward: 5'-caggtgtgccaaaggagcg-3'; reverse: 5'-cttcctaccaccccgaga-3'), and MX2 (forward: 5'-cttcagagacgcctcagtcg-3'; reverse: 5'-tgaagcagccaggaatagt-3'). All primers were synthesized by Sigma-Aldrich (St. Louis, MO). Relative expression of ISG15 and MX2 was determined using the $2^{-\Delta\Delta C_t}$ method with ACTB as the endogenous control. The stability of ACTB Ct values was verified across all sampling points. Gene expression data for ISG15 and MX2 were expressed as fold changes relative to d 14 for multiparous cows.

Statistical Analysis

A priori power analysis in G*Power (version 3.1.9; Faul et al., 2007) using the options of *t*-test for determining differences in 2 independent means with 2-tail test, $\alpha = 0.05$, power = 0.90, and effect size = 2.0 indicated that 7 pregnant cows or heifers per group was required to find a difference of 1 d on the day of PAG increase (d 20 vs. 21 and SD within each group = 0.5).

All statistical analyses were performed using SAS software (version 9.4). Statistical significance was set at $P \leq 0.05$ and tendency $0.05 < P \leq 0.10$. For gene expression (ISG15 and MX2), a linear mixed-effects model was used to account for repeated measurements within animals across days. Individual cows were treated as a random effect within their respective parity groups. Least squares means were used to compare groups and multiple comparisons were adjusted using Tukey's method. Data not normally distributed based on Shapiro-Wilk test were transformed into square root, logarithm, and rank. For P4 and PAG analyses, parity, day, and their interaction were included as fixed effects. To evaluate whether

the timing of PAG increase differed by parity group, a one-way ANOVA was performed. Pairwise comparisons were performed using Tukey's test. PROC GLM was used for area under the curve (AUC) analysis for circulating PAG. Correlation between average circulating P4 concentrations from d 4.5 to 12 and the day of PAG increase or the day PAG reached 8 ng/mL were evaluated using Pearson correlation. The effect of sire and sire by parity interaction were evaluated in preliminary analyses and found not significant; therefore, it was not included in the final model. Data are presented as mean $\pm$ SEM.

RESULTS

The relative abundance of cervical ISG and MX2 mRNA is presented in Figure 1A and 1B, respectively. There were a significant effect of time ($P < 0.001$) and a parity by time interaction ($P = 0.004$) for ISG15 expression. From d 14 to 16, no differences were detected among nulliparous, primiparous, and multiparous cows. On d 17, ISG15 expression was greater in primiparous and multiparous cows than in nulliparous heifers. On d 18, ISG15 expression was greater in primiparous compared with nulliparous. On d 20, primiparous cows had greater ISG15 compared with multiparous. When comparing ISG15 expression on different days within each parity category, ISG15 expression differed among days. In primiparous cows, ISG15 expression increased ($P < 0.05$) from d 14 to 15, peaked on d 17 and remained elevated on the subsequent days. In multiparous cows, ISG15 expression was greater ($P < 0.05$) on d 17 compared with d 14, peaked on d 18, and returned to levels similar to d 14 by d 20. For nulliparous heifers, d 16 was the only day that differed ($P < 0.05$) from d 14, indicating a weaker response compared with primiparous and multiparous cows. For MX2, there was only a main effect of day. Expression of cervical MX2 peaked on d 17 and decreased thereafter.

The profile of circulating PAG in the maternal circulation is shown in Figure 2. Basal concentrations of PAG were greater in primiparous and multiparous cows from d 14 to 18.5 compared with nulliparous heifers (Figure 2A). Concentrations of PAG were greater in nulliparous heifers compared with primiparous on d 20 to 21.5 and compared with multiparous cows between d 19.5 to 22 (Figure 2B). Overall, the first significant ($P \leq 0.05$) increase in the mean PAG concentration within each parity was on d 18.5 for nulliparous heifers, d 20.5 for primiparous cows, and d 21 for multiparous cows (Figure 2B). Considering the mean day of PAG increase in each animal, nulliparous heifers had earlier increase (d 18.9 ± 0.1) compared with both primiparous (d 20.0 ± 0.3 ;

$P = 0.0042$) and multiparous cows (d 20.6 ± 0.3 ; $P < 0.0001$; Figure 2C). The day on which cows reached the upper limit of the assay (8 ng/mL) differed by parity ($P = 0.0011$) with nulliparous heifers reaching this concentration earlier (d 23.4 ± 0.2) than multiparous cows (d 24.9 ± 0.3 ; $P = 0.0011$) and tending to be earlier than primiparous cows (d 24.3 ± 0.4 ; $P = 0.06$; Figure 2D). In addition, the time between initial increase in PAG and reaching the upper detection limit of the assay (8 ng/mL) did not differ among parity: 4.5 ± 0.3 d for nulliparous, 4.4 ± 0.4 d for primiparous, and 4.3 ± 0.2 d for multiparous cows (Figure 2E).

The profile of circulating PAG normalized to the initial increase in PAG (h 0) is shown in Figure 3A. There was only a significant effect ($P < 0.001$) of time with concentrations of PAG progressively increasing from the day of PAG increase through h 96 (Figure 3A). The AUC from h 0 to 96 was similar among groups (Figure 3B). The absolute increase in PAG relative to the day of PAG increase is presented Figure 3C. All parity groups had similar increase in concentrations of PAG over time.

Concentration of P4 increased after ovulation in all groups and remained elevated through the period evaluated (Figure 4A). Nulliparous heifers had greater concentrations of P4 compared with multiparous cows ($P = 0.0059$), whereas primiparous were not statistically different between either nulliparous or multiparous. On d 4.5, 10, and 12, nulliparous heifers had greater concentrations of P4 (0.60 ± 0.07 ; 3.37 ± 0.26 ; 4.15 ± 0.23 ng/mL) compared with multiparous (0.27 ± 0.06 ; 1.86 ± 0.28 ; 3.07 ± 0.23) and tended to be greater than primiparous (0.46 ± 0.11 ; 2.87 ± 0.35 ; 3.34 ± 0.31 , respectively). To further investigate the effect of circulating P4 on trophoblast differentiation, we focused on the period from d 4.5 to d 12 after AI. This interval was selected because studies have demonstrated that circulating P4 concentrations affects embryonic development in the first half of the estrous cycle but not later. Additionally, it precedes the typical onset of large amounts of conceptus-derived IFNT secretion. There was a negative correlation between the average circulating P4 from d 4.5 to 12 and the day of PAG increase ($r = -0.56$; $P = 0.0005$; Figure 5A). Similarly, there was a negative correlation between the average circulating concentrations of P4 during the same period and the day each cow reached a threshold of 8 ng/mL of PAG ($r = -0.44$; $P = 0.006$; Figure 5B). Cows with greater concentrations of P4 between d 4.5 and 12 had earlier increase in circulating PAG and reached the upper limit detection of the assay (8 ng/mL) sooner. There was no correlation between average circulating P4 and the interval between detection of PAG and reaching the assay threshold (Figure 5C).

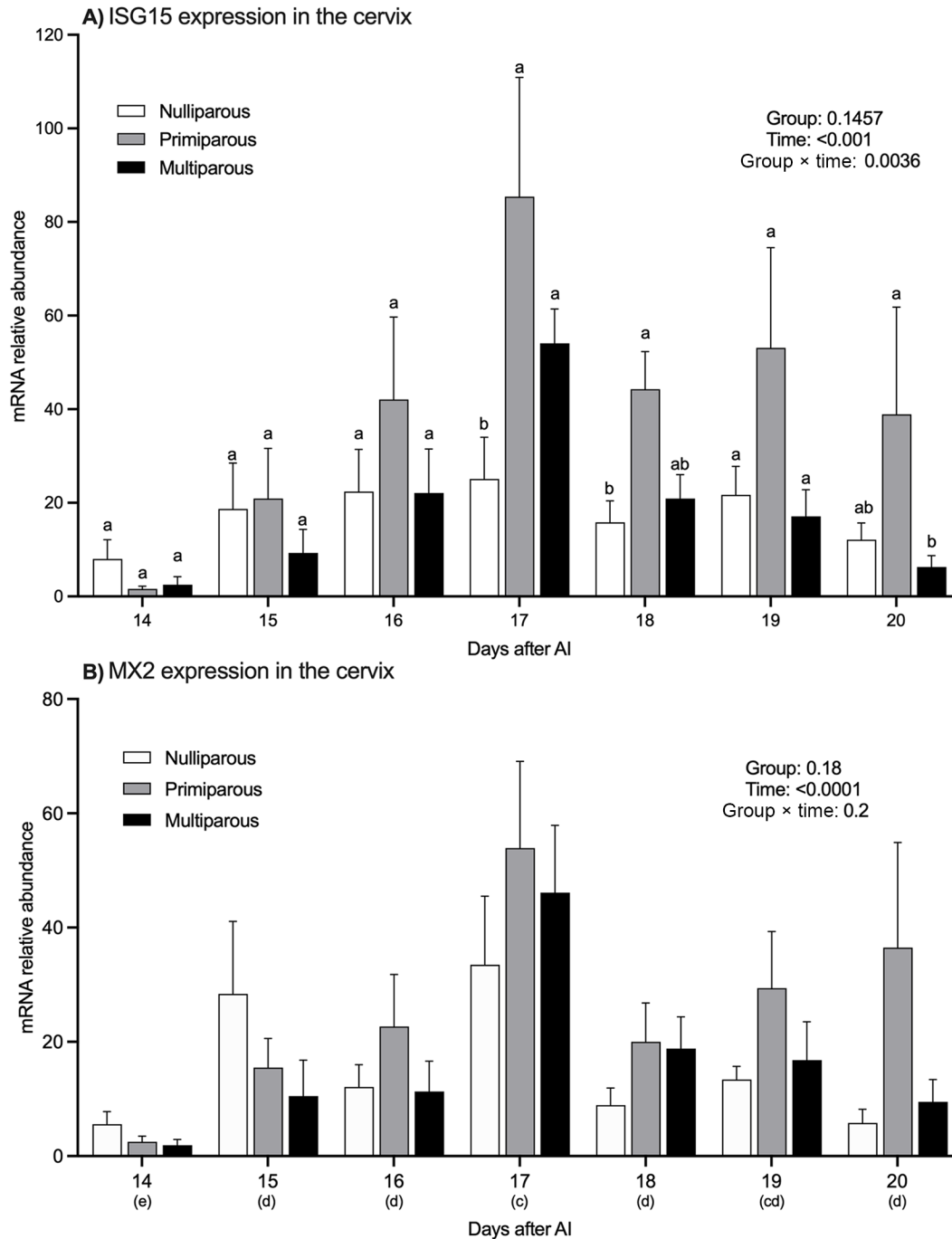


Figure 1. Abundance of mRNA for interferon-stimulated genes (ISG; [A] ISG15 and [B] MX2) in cervical epithelial cells from d 14 to 20 after TAI in pregnant dairy cattle categorized by parity (nulliparous, $n = 16$; primiparous, $n = 7$; multiparous, $n = 8$). Data are presented as mean $\pm$ SEM. Different lowercase letters indicate significant differences ($P \leq 0.05$) among parity within day (a,b) or among days (c–e).

DISCUSSION

Effective communication between the developing conceptus and the maternal system is fundamental for the successful establishment of pregnancy in cattle (Spencer et al., 2004b). This communication requires precise tem-

poral coordination between conceptus development and uterine receptivity, and disruptions in these processes are associated with pregnancy failure (Forde et al., 2011; Bridges et al., 2013; Lonergan and Forde, 2014). Heifers consistently have greater pregnancies per AI (Pursley et al., 1997; Sartori et al., 2002, 2009) and per embryo

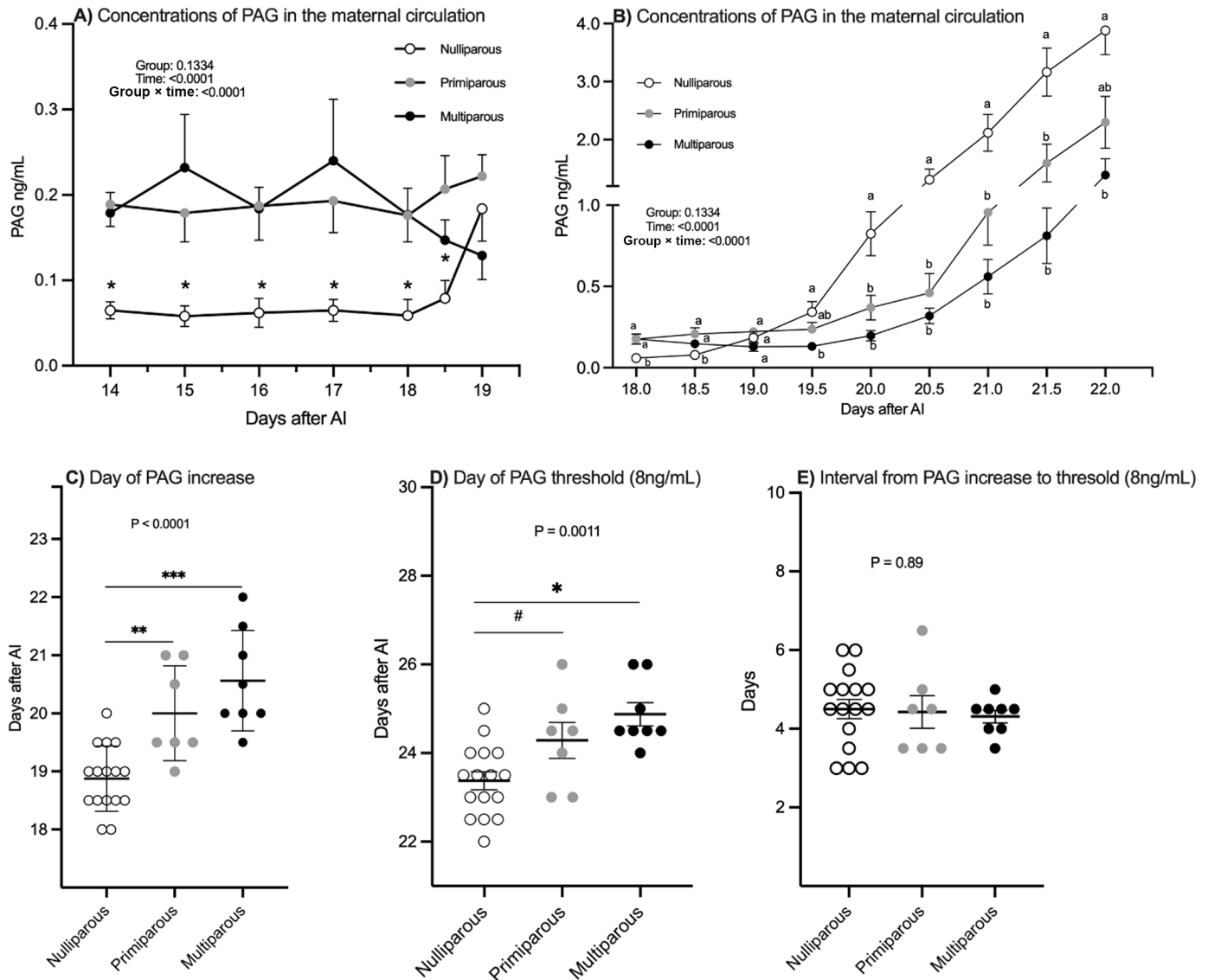


Figure 2. Concentrations of pregnancy-associated glycoproteins (PAG) in pregnant cattle categorized by parity (nulliparous, $n = 16$; primiparous, $n = 7$; multiparous, $n = 8$) from d 14 to 19 (A) and d 18 to 22 (B). The day of PAG increase above the baseline (C), and the day it reaches the upper detection limit of the assay (8 ng/mL; D) are shown. (E) The interval from initial PAG increase to reaching 8 ng/mL. In panels C, D, and E, circles indicate individual cow data. Data presented as mean $\pm$ SEM. Different lowercase letters (a,b) indicate significant differences ($P \leq 0.05$) among parity within each day. In panel A, * indicates significant differences between nulliparous and both primiparous and multiparous cows. In panels C and D, # indicates $P < 0.1$, * indicates $P < 0.05$, ** indicates $P < 0.01$, and *** indicates $P < 0.0001$.

transfer (El Azzi et al., 2023) compared with multiparous cows; yet, the biological mechanisms underlying this advantage remain poorly understood. The present study provides new insight by concurrently evaluating the dynamics of IFNT–ISG signaling, early placentation via PAG concentrations in maternal circulation, and circulating P4 in heifers, primiparous cows, and multiparous cows under tightly controlled conditions. This integrated approach allowed us to isolate parity effects on peri-implantation development more precisely than previous studies.

Contrary to our hypothesis, ISG15 expression in cervical epithelial cells did not differ between parities, indicating similar maternal responsiveness to IFNT during the maternal recognition of pregnancy. However, consistent with Middleton et al. (2022), nulliparous heifers showed an earlier rise in circulating PAG compared with both primiparous and multiparous cows. Differently than previous studies, our study evaluated not only circulating PAG concentrations but also ISG from nulliparous heifers and primiparous and multiparous cows. The strength of our experimental design—including identical synchroniza-

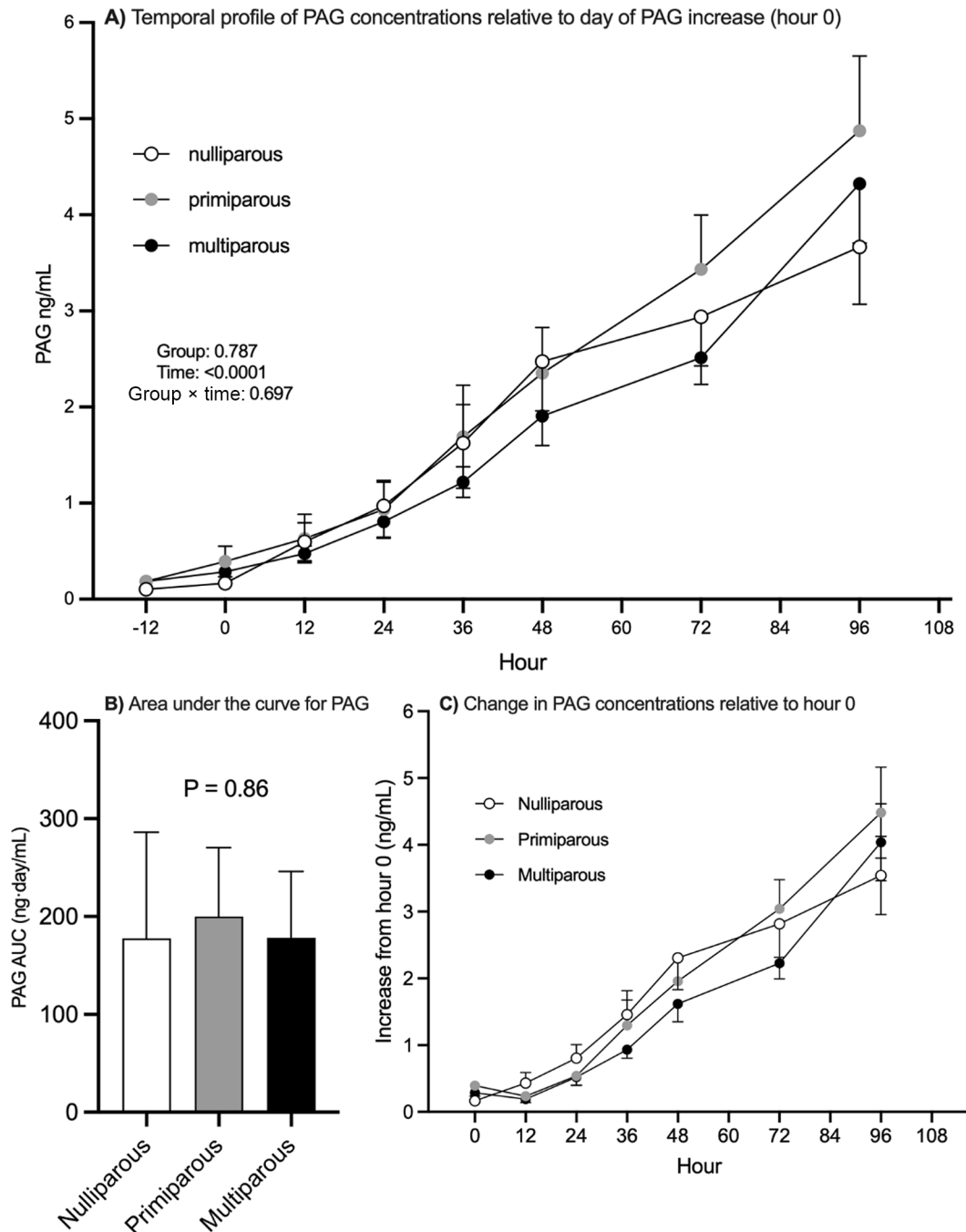


Figure 3. Profile of concentrations of pregnancy-associated glycoprotein (PAG) in maternal circulation relative to the day of PAG increase (h 0) in pregnant cattle categorized by parity (nulliparous, n = 16; primiparous, n = 7; multiparous, n = 8; A). (B) Area under the curve (AUC) for PAG concentrations from h 0 to 96. (C) Change in PAG concentrations relative to h 0 across time for each parity group. Data are presented as mean $\pm$ SEM.

tion protocols across parities, use of frozen conventional semen from similar sires, confirmed ovulation within 24 h of TAI, and circulation of PAG concentrations every 12 h from d 18 to 24 after AI—reduces confounding variation and provides sound evidence that parity alters the timing of peri-implantation development. Importantly,

our results provide new evidence that the greater overall circulating PAG observed in nulliparous heifers compared with primiparous cows from d 20 to 21 and with multiparous cows from d 19.5 to 22 likely reflect earlier secretion of PAG rather than a faster rate of PAG production, providing new insight into how parity influences

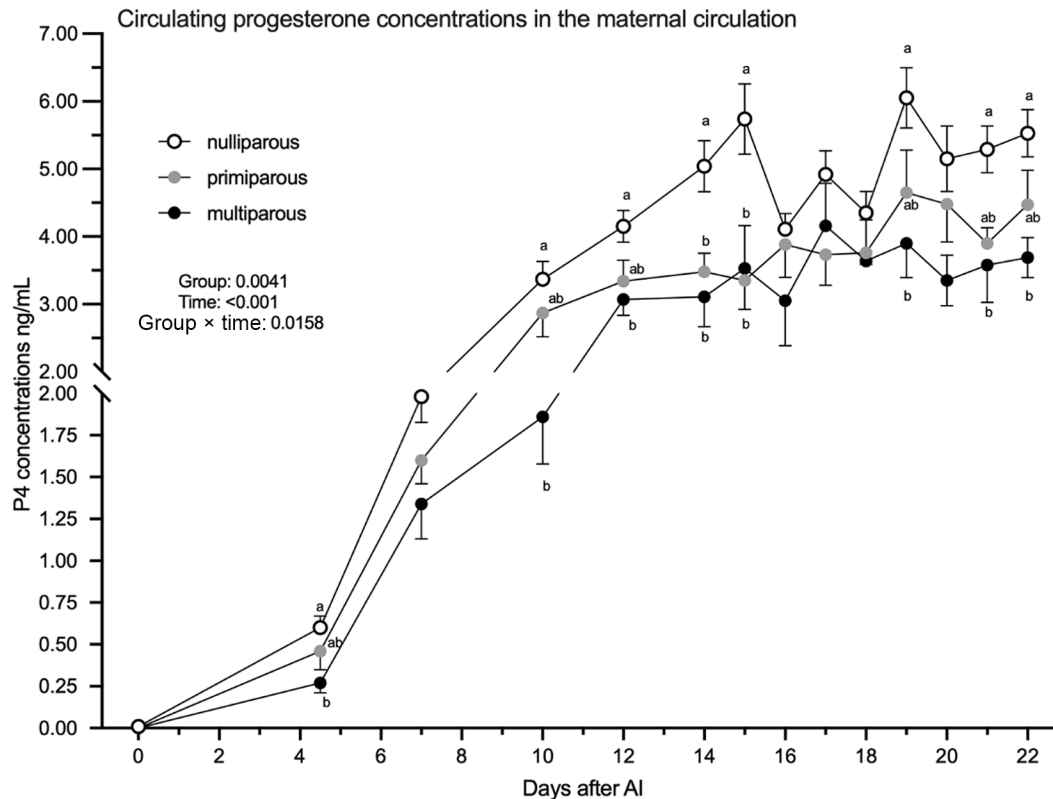


Figure 4. Circulating concentrations of progesterone (P4) in pregnant cattle categorized by parity (nulliparous, n = 16; primiparous, n = 7; multiparous, n = 8). Different lowercase letters (a,b) indicate significant differences ($P \leq 0.05$) among parity within each day. Data are presented as mean $\pm$ SEM.

early embryonic development in the peri-implantation period.

The temporal expression of ISG15 and MX2 in cervical epithelial cells reflects the dynamic communication between the developing conceptus and the maternal system during early pregnancy (Talukder et al., 2023, 2024; Feltrin et al., 2025). Domingues et al. (2024) demonstrated that ISG15 expression in the cervix is markedly greater in pregnant cows compared with nonpregnant cows. In endometrial cells, the expression of ISG correlates to the amount of IFNT, at least in vitro (Kim et al., 2013; Chaney et al., 2021; Talukder et al., 2023). However, the assessment of ISG in the present study does not allow discernment between the amount of IFNT secreted by the embryo and the maternal responsiveness to IFNT. Nevertheless, the mRNA profile of ISG15 and MX2 were maximal around d 17, which is consistent with the period of maximal IFNT secretion from the elongating conceptus (Hansen et al., 2017; Imakawa et al., 2017; Sheikh et al., 2018). Cervical abundance of ISG15 in pregnant cows has been reported as early as d 14, peaking on d 16 after TAI (Mendina et al., 2025). Notably, the cervix is a highly responsive tissue to IFNT during maternal recognition of pregnancy, yielding greater confidence to

determine pregnancy status and assessing the maternal response to IFNT over time.

A novel finding in the present study is that the timing of key events in early embryonic development differs according to parity in dairy cattle. Notably, nulliparous heifers had less mRNA abundance of ISG15 on d 17 and 18 and earlier increase in circulating PAG compared with cows suggesting hastened conceptus development compared with lactating cows. Embryonic secretion of IFNT decreases as the trophoblast differentiates and fuses with uterine luminal epithelial cells due to epigenetic modifications that remodel chromatin structure at the IFNT locus, leading to suppression of IFNT expression (Sakurai et al., 2012). This process is not abrupt but rather gradual as the IFNT expression persists in areas where the trophoblast has not yet attached but decrease progressively in regions undergoing attachment (Guillomot et al., 1990; Imakawa et al., 2017). Therefore, the decreased ISG15 in nulliparous heifers appears to be due to localized suppression of IFNT in regions of trophoblast differentiation and attachment to the uterine endometrium. This is consistent with the earlier increase in PAG in heifers compared with primiparous and multiparous cows and reflects hastened embryonic development, trophoblast differentiation, and

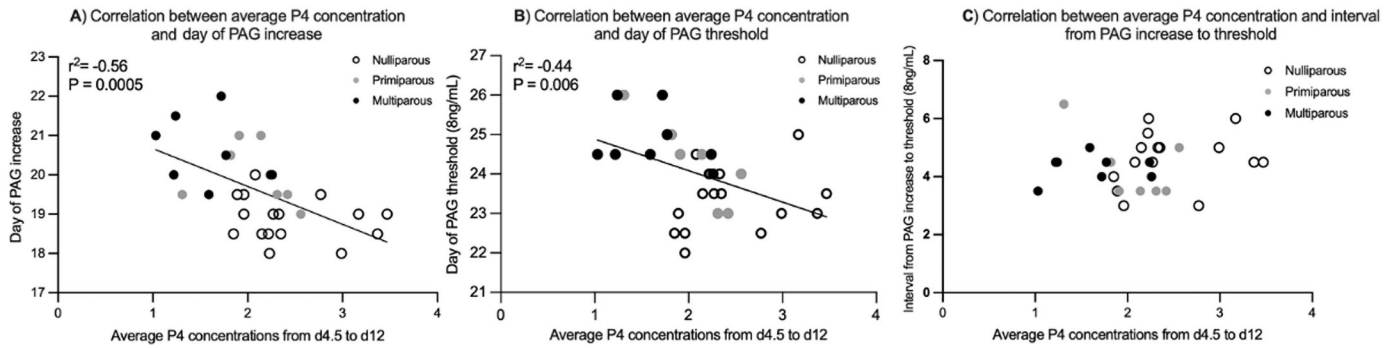


Figure 5. Pearson correlation analyses between average circulating progesterone (P4) from d 4.5 to 12 and (A) the day of increase in circulating of pregnancy-associated glycoprotein (PAG) secretion, (B) the day concentrations of PAG reached the upper limit detection of the assay (8 ng/mL), and (C) interval from PAG increase to threshold.

implantation. These findings are similar to prior research showing that systemic PAG is affected by parity in both dairy (Ricci et al., 2015; Middleton et al., 2022) and beef cattle (Lobago et al., 2009).

In addition, the weaker ISG15 response observed in nulliparous heifers compared with primiparous and multiparous cows may reflect differences in prior exposure of maternal tissues to interferons. Recent evidence indicates that cells previously exposed to interferons can develop a memory-like state, resulting in faster and stronger transcriptional responses upon restimulation (Kamada et al., 2018). Thus, prior exposure to conceptus-derived IFNT during a previous pregnancy may have primed cervical epithelial cells in parous cows to mount a more robust ISG15 response, whereas nulliparous heifers encountering IFNT for the first time may exhibit a comparatively attenuated response.

The hastened embryonic development in heifers is likely multifactorial, as several conditions are known to affect embryonic survival: oocyte competence, metabolic challenges, and uterine health and function, among others (Santos et al., 2004; Wiltbank et al., 2016). Owing the critical role for luteal P4 in regulating uterine function (Carter et al., 2008; Clemente et al., 2009; Forde et al., 2009), the greater circulating P4 in heifers likely plays a role in enhancing embryonic development. For instance, P4 supplementation seems to modulate the composition of the histotroph in cattle, resulting in faster embryonic elongation (Carter et al., 2008; Forde et al., 2009). In the present study, multiparous cows had decreased circulating P4 compared with nulliparous heifers, which can be explained by the increased hepatic blood flow of lactating dairy cows due to greater DMI to support lactation (Sangsrivong et al., 2002; Sartori et al., 2006; Wiltbank et al., 2014). In contrast, greater P4 in heifers are likely to enhance endometrial function creating a uterine environment that supports better conceptus elongation

and earlier trophoblast differentiation (Clemente et al., 2009; Tyree and Stenhouse, 2025). This is further supported by the negative correlation between P4 from d 4.5 to d 12 and the day of PAG increase, indicating that animals with greater P4 during this critical period have enhanced embryonic development and earlier onset of PAG secretion. In contrast, P4 was not correlated with the time needed to reach the PAG threshold, indicating that although P4 hastens the onset of trophoblast differentiation, it does not appear to influence its subsequent development. Another potential mechanism related to the earlier attachment in heifers is related to uterine size. It is plausible that the smaller uterus in heifers facilitates the contact between the elongating conceptus and the endometrium, thereby hastening the embryonic attachment to the endometrium (Baez et al., 2016). Although the uterine environment plays a major role in determining conceptus development (Moraes et al., 2018), oocyte competence may also play a role in the hastened embryonic development in heifers. As discussed by Leroy et al. (2008), metabolic stress associated with lactation can compromise oocyte quality by altering the follicular environment, which negatively affects the oocyte's ability to support subsequent development. Similarly, Rizos et al. (2005) highlighted that oocyte quality is determined by a combination of factors including metabolic status, physiological history, and management, emphasizing that developmental competence is influenced by more than just lactational status or parity. This suggests that embryos from heifers, which are less affected by the metabolic burden of lactation, may be more capable of initiating development earlier when the uterine environment becomes supportive. Lastly, it is unlikely that the observed differences in embryonic development and pregnancy-associated markers across parities are due to genetic differences among animals (Diskin et al., 2015; Sigdel et al., 2022). The same individual with high fertility

as a nulliparous heifer often shows reduced fertility as a primiparous or multiparous cow (Pryce et al., 2004; Walsh et al., 2011). This pattern suggests that physiological changes related to lactation and increasing parity, rather than genetic merit alone, are primarily responsible for the decline in reproductive efficiency in older cows compared with nulliparous heifers (Butler, 2003; Walsh et al., 2011).

An interesting finding in the present study is related to the basal circulating concentrations of PAG in heifers compared with cows. Before the onset of the PAG increase, nulliparous heifers had consistently lower basal concentrations of PAG from d 14 to 18.5 after TAI compared with cows. This disparity is unlikely to reflect conceptus secretion of PAG before d 18 in cows but rather results from previous pregnancies. Circulating PAG is maximal before labor (Zoli et al., 1992; Monteiro et al., 2024) and decreases gradually after parturition but remains detectable at low concentrations for several weeks postpartum (Green et al., 2005). A similar pattern occurs following pregnancy loss (Domingues et al., 2023); PAG decreases more rapidly in the first few days but still takes more than a week to return to baseline (Giordano et al., 2012). Thus, a low residual PAG from previous gestations likely explains the greater baseline observed in lactating cows. Although overall little is known about the metabolic clearance of PAG, the greater liver metabolism due to increased DMI during lactation (Sangsritavong et al., 2002) does not appear to affect circulating concentrations of heifers versus cows. This is further supported by the similar increase in PAG after normalization to the onset of detection of PAG according to parity status. Therefore, after the onset of PAG secretion, differences in circulating PAG are primarily associated with conceptus development or secretion rather than metabolic clearance.

CONCLUSIONS

Our data demonstrated that embryonic development is hastened in heifers suggesting earlier trophoblast differentiation with acquisition of PAG secretory capacity compared with primiparous and multiparous lactating cows. This advancement was associated with greater circulating P4 from d 4.5 to 12 after TAI, suggesting that parity-related differences in hormonal environment might contribute to hastened embryonic development in heifers. In addition, differences in cervical epithelial cell ISG15 expression among parities indicate that parity influences the temporal dynamics of IFNT-mediated conceptus–maternal signaling during early pregnancy. Notably, despite the earlier increase in circulating PAG concentrations in nulliparous heifers, no earlier increase in ISG15 or MX2 expression was detected, suggesting

that parity affects the coordination, rather than the onset, of IFNT signaling and trophoblast differentiation.

NOTES

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Nonstandard abbreviations used: AI = artificial insemination; AUC = area under the curve; IFNT = interferon-tau; ISG = interferon-stimulated genes; P4 = progesterone; PAG = pregnancy-associated glycoprotein; qRT-PCR = quantitative real-time PCR; TAI = timed AI.

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